PG No. 10 S_4 $\bar{4}$ [tetragonal]

Table 1: Representation matrices

Irrep.				
A	$1: \begin{bmatrix} 1 \end{bmatrix}$	$2_{001}: \begin{bmatrix} 1 \end{bmatrix}$	$-4^{+}_{001}: \begin{bmatrix} 1 \end{bmatrix}$	$-4^{-}_{001}: \begin{bmatrix} 1 \end{bmatrix}$
B	$1: \begin{bmatrix} 1 \end{bmatrix}$	$2_{001}: \begin{bmatrix} 1 \end{bmatrix}$	$-4^{+}_{001}: \begin{bmatrix} -1 \end{bmatrix}$	$-4^{-}_{001}: \begin{bmatrix} -1 \end{bmatrix}$
E	$1: \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$2_{001}: \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	$-4^{+}_{001}: \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$-4^{-}_{001}: \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$